

Solution Requirements

Date	26 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely log in to access the Smart Lender system.	High
2	Authorization Levels	Admin manages the system, while loan officers/applicants access only permitted features.	High
3	External Interfaces	The system accepts applicant information through a Flask web interface and uses the trained ML model for prediction.	High
4	Transaction Processing	Applicant details are processed, validated, and loan eligibility is predicted instantly.	High
5	Reporting	Displays loan approval/rejection result and prediction summary.	Medium
6	Business Rules	Loan eligibility is determined based on applicant income, credit history, education, loan amount, and other features using the trained XG Boost model	High
7	Compliance to Laws or Regulations	Applicant data is stored securely and privacy is maintained.	Medium
8	Other	System stores prediction results for future analysis.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric/ Acceptance Criteria
1	Performance & Speed	Predict loan eligibility quickly.	Prediction in less than 3 seconds
2	Scalability	Support many applicants without performance issues.	Supports 500+ users

3	Security & Data Privacy	Protect applicant information using secure storage and authentication.	User data remains confidential
4	Reliability & Availability	System should work without failures.	99% availability
5	Usability & Accessibility	Easy-to-use web interface for all users.	User-friendly interface
6	Other	Maintainable and portable across Windows/Linux.	Easy updates and deployment